Innovation Statement

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An overwhelming majority of cancer deaths is caused by drug resistance in patients receiving targeted therapies [1, 2]. Contemporary approaches like DRN-CDR [12] and MSDRP [13] retuan a probability score from multi-omics feature vectors. When the fetures conflict, the learned weights resolve it silently without any justification. Three recent developments pave the way for a better alternative: (1) Model Context Protocol (MCP), standardized in 2024, enables LLMs to query structured databases using schema-validated tool calls instead of free-text. (2) LLMs now follow multi-step tool-use protocols reliably for structured reasoning chains [7, 8]. (3) GDSC2 and CCLE/DepMap now provide multi-omics profiles at the required scale and quality for multi-modal reasoning [11]. Published frameworks for tumor boards document a hierarchy of evidence in treatment decisions (DNA -level findings designated as primary determinants when they conflict with expression data [3, 4, 10]). It remains crucial to see whether encoding the hierarchy computationally improves fidelity and accountability. We introduce 5 innovations that have not appeared individually or in combination in today's literature.

<u>A formalized biological Hierarchy of Truth governing inter-agent conflict resolution.</u> The use of multi-agent debate systems in oncology [3, 4] have shown that any structured disagreement among specialized AI agents show better clinical outcomes compared to single agent systems. However, no existing literature has a transparent resolution strategy or a testable logic. We design 4 specialized agents (Genomics, Transcriptomics, Pathways and Pharmacology), each analyzing a case independently before exchanging evidence. **We introduce a 5-tier axiom hierarchy (T1 Structural Primacy → T2 Transcriptional Gate → T3 Pathway Bypass → T4 Pharmacological Prior → T5 Statistical Consensus) as executable rules, making the process of resolution fully reproducible and biologically grounded.** Where a probabilistic model returns `P(SENSITIVE) = 0.61` with no justification, this study surfaces the same conflict explicitly: Genomics agent returns SENSITIVE (BRAF V600E, T1), Transcriptomics agent returns RESISTANT (BRAF silenced, T2), T1 overrides T2, final verdict will be SENSITIVE and dissent gets logged.

<u>The Model Context Protocol as a structural anti-hallucination layer.</u> The framewrk separates two communication layers: agents query databased via MCP (Agent-to-Tool), then debate each other's evidences via an Agent-to-Agent protocol. Existing strategies pass input data to the agents as free-text summaries and rely on model self-restraint to prevent any hallucinations. **By routing the respective agent data request through MCP-validated tool calls that return schema-validated JSON [7, 8], we prevent an agent from citing any mutations that aren't in the database**. For example, an agent cannot assert a BRAF V600E mutation if the genomics MCP server does not return one.

<u>A debate trace and dissent log as a clinical accountability artefact.</u> Every prediction produced

by the system has a structured record that contains information of which of the agents agree, which were overridden by which axiom tier, and at which debate round. Agents whose verdict was superseded by a higher axiom tier retain the dissent in the final output to show how minority opinions are recorded in the case of real tumor board discussions. **This debate trace is a novel artefact, enabling the oversight of clinicians and researchers and post-hoc audit of every AI recommendation [3, 10].**

A novel synthesis of multi-agent debate, structured tool access and biological axiom logic. Existing literature combines at most two of these elements. EvoMDT [3] uses a debate but no axiom logic, BioAgents [9] use tool-anchored reasoning without conflict resolution, DRN-CDR [12] integrates multiomics data without transparency. **This framework uses structured debates, MCP-enforced access to data, explicit biological axiom hierarchy, all simultaneously, and in doing so creates a system that does not previously exist.**

Purpose-built infrastructure. This system is designed using unique evaluation resources: four MCP servers wrapping GDSC2, CCLE/DepMap, KEGG and ChEMBL using schema-validated APIs to prevent hallucinations. We use 40 GDSC2 cases stratified across the five axiom tiers for stress-testing. The system is LLM-agnostic, supporting Gemini Flash, Qwen and Gemma with responses cached in SQLite. **This design enables efficient comparison of different LLMs on a multi-omics reasoning task.**

Overall, these five innovations shift how we evaluate oncology AI from predictive accuracy alone to *reasoning quality*, in terms of hallucination rate, biological faithfulness of debate traces and axiom invocation patterns. We ultimately have an open-source, audit-ready framework for multi-modal biological reasoning that is generalizable beyond drug resistance to other domains requiring principled reconciliation of evidence conflicts.

Reflection

Going through the sample grants on the NIH website made it clear on how to structure an innovation statement. These sections typically read as a single continuous argument, moving from addressing the research gap to novel contributions to implications. Each novelty is underlined to let a reviewer scan the novelties without breaking the prose as subsections.

Feedback

One crucial feedback provided was to lead with the gap rather than the enabling advances. This is to make reviewers understand that this is a real problem that needs to be addressed before we can see how to solve it. The BRAF V600E example was also an add-on to provide an illustration of the axiom hierarchy for easier understanding.

References

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